EEEN10131 Digital System Design I

Laboratory 1 – Combinatorial Logic

1. Instructions:

- Read the introductions and make necessary preparations.
- Complete the tasks in Section 4 (Tasks 1-8).
- Submit the work from page 9 onwards in a single pdf file to the Blackboard for lab1 before the deadline (**see Unit Delivery Plan in Blackboard for the deadline**).

2. Introduction to NI myDAQ and NI ELVISmx software instruments

Getting Started

You can use a computer in university clusters (with NI ELVISmx installed) to complete the tasks of Laboratory 1 before your scheduled attendance during which you are only required to demonstrate your circuit. Alternatively, if you decide to use your own PC, you need to install a few software applications before you can use your NI myDAQ. Instructions for this process are given separately in the file of Preparation for Labs.

Configuring NI myDAQ

Once you have successfully installed the software suite, plug the myDAQ to your computer via USB. The computer should recognise the myDAQ and the NI ELVISmx Instrument Launcher should appear.

Check whether myDAQ is successfully connected to the PC.

(1) Once the myDAQ has been plugged in to the PC via the USB cable, open the **National Instruments Measurement and Automation Explorer (NI MAX)** . You can find this by navigating to **Start>> All Programs>>National Instruments>> NI ELVISmx for NI ELVIS & NI myDAQ>> NI ELVISmx Instrument Launcher.** Once NI ELVISmx Instrument Launcher opens, navigate to 'Resources' tab. You will find NI MAX here (see Figure 1).

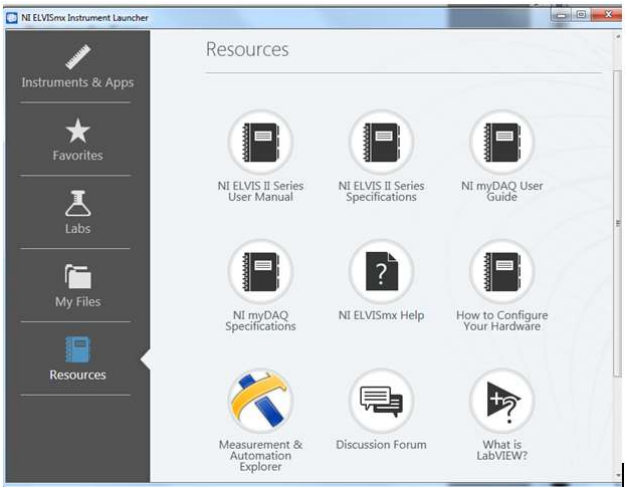


Figure 1: How to navigate to NI MAX

(2) Navigate to Devices and Interfaces Section. Expand the tree under Devices and Interfaces.
(3) Check the myDAQ is listed and then select it and click Self Test to ensure it is connected to the PC correctly. Make sure the name of the device is “myDAQ1”.

Important:

Ensure that the name of the device set to “myDAQ1”. Otherwise the program will not recognise the device and will not run.

Ensure that the black 5 V switch on the breadboard is at the ‘ON’ position. Otherwise the circuit will not receive power and will not give the correct output.

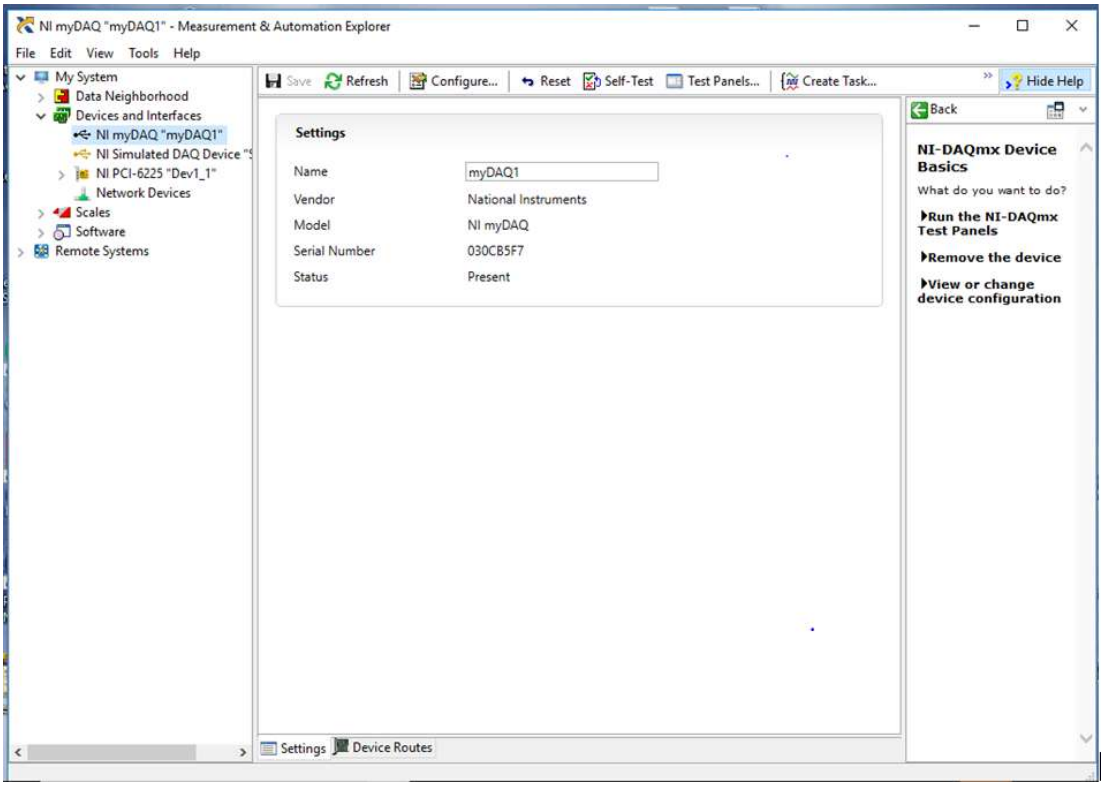


Figure 2: NI Measurement and Automation Explorer window

NI ELVISmx Instrument Launcher

NI ELVISmx Instrument Launcher provides access to NI ELVISmx software instruments. NI ELVISmx software instruments are created using LabVIEW source code and together with your myDAQ, they provide you with functionality of a range of laboratory equipment.

When you plug in NI myDAQ, NI ELVISmx Instrument Launcher should open automatically. You can also open it by navigating to **Start>> All Programs>>National Instruments>> NI ELVISmx for NI ELVIS & NI myDAQ>> NI ELVISmx Instrument Launcher.**

In this lab we will be using Digital Multimeter (DMM), Digital Reader and Digital Writer. Let’s explore the functionality of each of these software instruments in a little more depth.

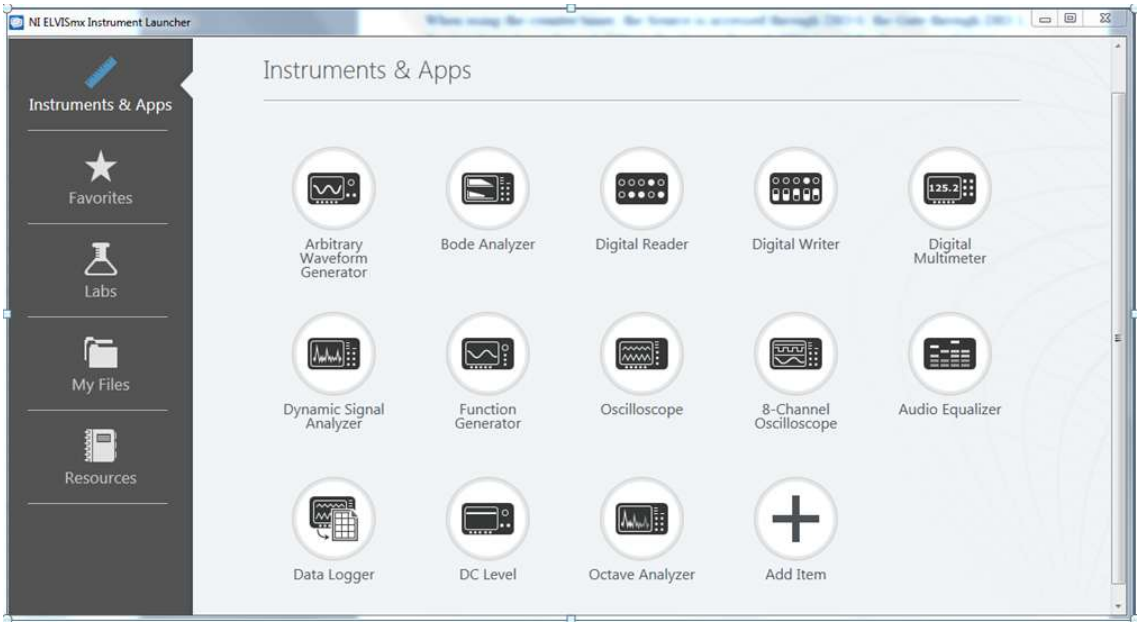


Figure 3: software instruments available for NI myDAQ in NI ELVISmx Instrument Launcher

Digital Multimeter (DMM)

This instrument can measure voltage (AC and DC), current (AC and DC) and resistance. It can also perform diode tests and audible continuity tests. In this lab we will use the DMM to perform voltage measurements.

How to perform a voltage measurement

You have to connect the two probes in parallel with the component/two points in the circuit that you want to measure the voltage across.

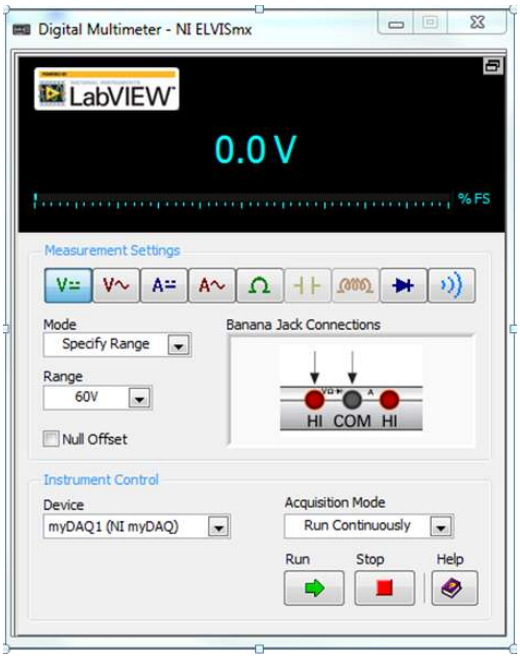


Figure 4: DMM. Notice the Banana Jack connections required for the measurement you are performing is shown on the DMM.



Figure 5: How the probes must be connected to measure the voltage

Digital Writer

Digital Writer writes to the NI myDAQ digital lines (DIO) that are input lines to the circuit in concern. You can manually create a pattern or use predefined patterns such as ramp, toggle or walking 1s. You can either continuously output a pattern or just perform a single write.

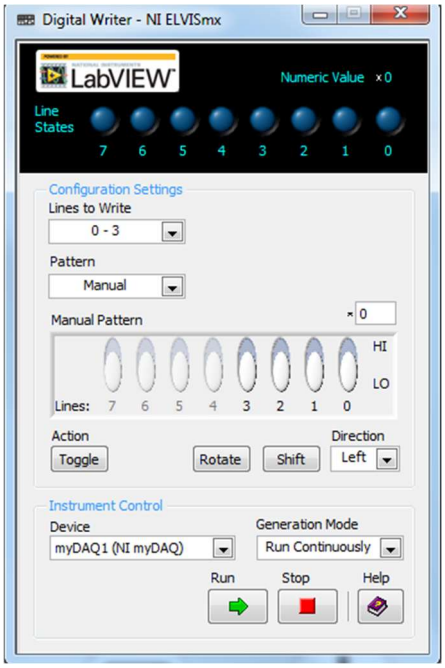


Figure 6: NI ELVISmx Digital Writer

Digital Reader

Digital Reader reads from the NI myDAQ digital lines (DIO) that are output lines of the circuit in concern. You can perform the reading continuously or as a single reading.



Figure 7: NI ELVISmx Digital Reader

3. Introduction to DIGILENT myProto board

DIGILENT myProto breadboard can be attached directly to the NI myDAQ. Signals from/to the circuit built on the breadboard can be directly connected to the myDAQ signals using jumper wires. Signal names on the breadboard indicate which myDAQ signals are connected to which breadboard pins.

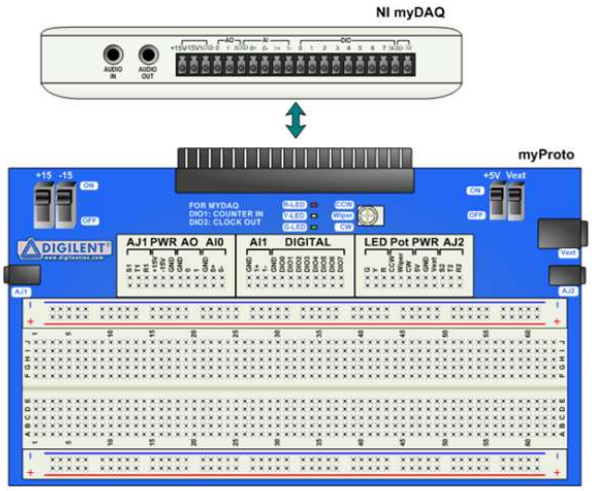


Figure 8: Connecting DIGILENT myProto board to the NI myDAQ (image source: DIGILENT myProto board reference Manual)

The breadboard has holes to allow components to be connected together by wires (single core wires are used for circuit building on breadboard) /jumper cables that are inserted into the holes.

The long central area of the breadboard comprises 63 rows of holes. Each row has 2 groups of 5 holes. The 5 holes in a group are connected together underneath the board via copper tracks. Thus two pins of the same component must not be connected to two holes of the same row and same column, unless you want to short circuit those two pins.

There are columns labelled '+' and '-'. The holes in any one of these columns are connected together underneath via copper tracks. These are used to provide shared power supplies to the circuit that is to be built. For these laboratory session the supplies will be 5V for '+' column and 0V (Ground-GND) for '–' column.

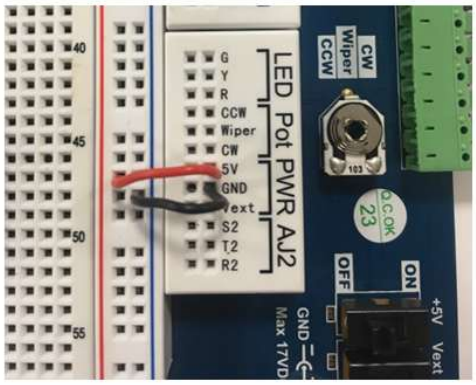


Figure 9: How to connect 5 V to + rails and Ground to – rail

How to get the pin numbering for a Dual-In-Line (DIL) Package right

The notch on the package indicates the top side of the package. If you hold the DIL package in such a way that the notch is at the top, the left leads are numbered from top to bottom and the right leads are numbered from bottom to top. In other words, pin numbering is anticlockwise.

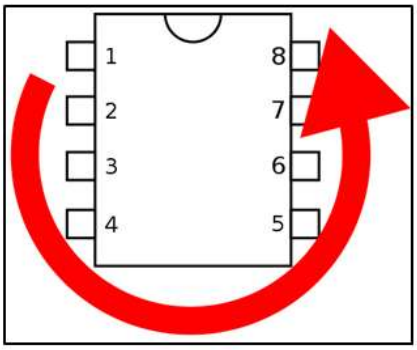


Figure 10: Pin-Numbering of convention for a DIL package (image source: Wikipedia)

How to place a component with a dual in line package on a breadboard

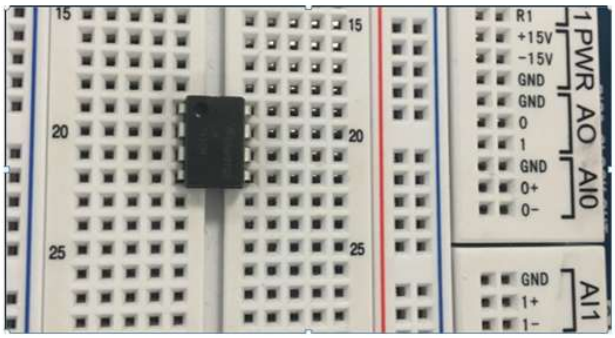


Figure 11: How to place a DIL package on a bread-board. The picture shows an LM741 op-amp.

On a breadboard every row of a specific column are connected together. Therefore, left set of leads and the right set of leads of a DIL packaged component must be placed on two separate columns of a breadboard as shown in the above image.

Essential Lab preparation

You will be using logic chips to build simple combinatorial logic circuits on your breadboard. You will appreciate that building even a simple circuit on a breadboard using wires consumes a lot of area. Modern digital systems contain thousands of logic components placed and routed on very small areas of silicon. Therefore, to design modern digital systems it is essential to use computer aided software. This design approach is called EDA-Electronic Design Automation.

You will be measuring the output from the logic circuits using DMM (analogue measurements) and Digital Reader (digital measurements). Therefore, you will see that logic low and high are represented by a range of voltages; not precisely 0 V and 5 V respectively.

What you will need to complete this laboratory exercise

- A PC which has the complete NI myDAQ suite installed
- The complete NI myDAQ kit
- The laboratory specific component kit provided for you

What you need to know before starting the laboratory

Two integrated circuits comprising 4x2-input NAND gates and 2x4-input NAND gates are provided for you to implement the logic circuit corresponding to the assigned truth table. Before starting the lab you should understand how,

- Canonical sum of products expressions are derived
- Reduced logic expressions are obtained from the derived canonical sum of products
- Reduced logic translates into a circuit diagram (here only using 2 input and/or 4 input NAND gates)

Example: [Note: This is an example. Use your allocated Zout for your Lab1 work. See Task 7 for your allocated Zout]

Follow the example below carefully and ensure that you understand how logic expressions are derived at every stage.

Below is a truth table for a 3 input combinatorial logic circuit

A	B	C	Z
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0

The canonical sum of products expression is given by:

$$Z = \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}C$$

From this the reduced logic expression is given by:

$$Z = \bar{A}B + A\bar{B}C$$

Using De Morgan's law, this can be expressed in such a way that only NAND gates are needed to implement the logic circuit:

$$Z = \overline{\overline{\bar{A}B} \cdot \overline{A\bar{B}C}}$$

As only 2 input and 4 input NAND gates are provided, the logic diagram can be drawn as:

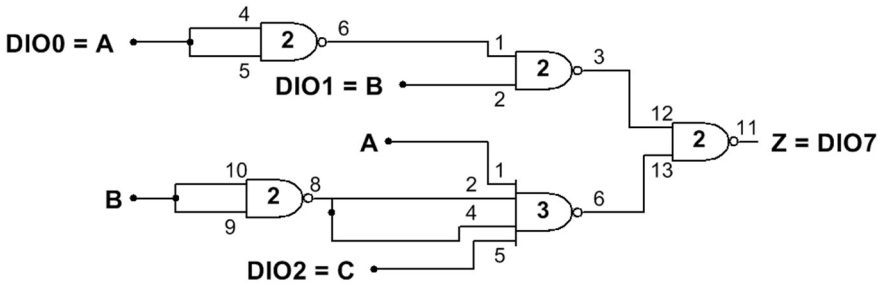


Figure 12: The logic circuit to be built. Note all the data that is included in the diagram. When you draw your circuit later in the laboratory, you should include all these details to gain full marks.

Note that,

- An inverse function is realised using a single two input NAND gate
  For instance, $\overline{A \cdot A} = \bar{A} + \bar{A} = \bar{A}$
- The 4th input of the 4-input NAND gate is not just left 'floating' (without any connection).
- The data that must be included to give a full description of how to construct the circuit are,
  a) Identification of the chips that are used for each gate (in this case, "2" is SN74AC00N and "3" is 74HC20N).
  b) Pin numbers on the integrated circuits for the inputs and outputs of each gate
  c) Identification of the connections to the ELVIS board.

(When you draw your circuit later in the laboratory, you should include all the above details to gain full marks)

Important:

Accuracy of measurements

In some tasks, you are asked to give the measurements to mV accuracy. This essentially encourages you to consider which of the digits are significant. You will notice that some of the less significant digits change continuously. This is due to 'noise' of the system. Should these changing digits be included in your recorded measurements? They should not be since they do not add any more accuracy to your measurements.

For instance, a voltage reading of 4.325789 V can be recorded to mV accuracy as 4.326 V or 4326 mV.

Recording Measurements

It is very important to record what you actually observe. This might not necessarily be the outcome that you expect. In this laboratory, it is possible for some chips to "misbehave". If you suspect that this is the case, then you should say how the behaviour is different from your expectations in your responses on the sheet.

4. EEEN10026-Lab1: Lab Tasks

After completion, submit the work in page 9 upto the end page in a single pdf file to Blackboard.

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First Name	**PeaceofGod Itunuoluwa**
Student ID	**11377069**
Note:	

PART A: Behaviour of a 2-input OR gate using SN74LS32N

In this part of the lab we will be using SN74LS32N chip by Texas Instruments. It consists of four 2-input OR gates and the chip is powered by 5 V and 0 V (GND) supply on the breadboard. The SN74LS32N uses TTL (Transistor-Transistor Technology), as suggested by the “LS” in the name. TTL is historically important but is now less prolific than CMOS technology.

We will use the 5 V power supply from the myDAQ for logic HIGH inputs and Ground (0 V) from the myDAQ for logic LOW inputs. We will be measuring the output from the OR gate using the DMM to get an analogue measurement (the exact voltage output from the OR gate).

Important: You should double check that you have connected the correct pins to the power supply before powering on the myDAQ. Otherwise, you could damage the chips.

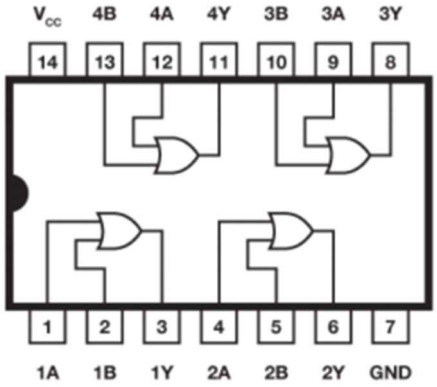


Figure 13: Pinout for SN74LS32N (Image source: datasheetdir.com)

(Build the circuit on the breadboard before switching on the 5 V power supply switch on the breadboard)

- Place the SN74LS32N chip on the breadboard. Connect the VCC pin to 5V and GND pin Ground of myDAQ.
- There are 4 OR gates on this chip. We will use the OR gate number 1. A (pin 1) and B (pin 2) pins are inputs while Y pin (pin 3) is the output of the OR gate 1. The pin numbering of a DIL package is explained in more detail in the previous sect

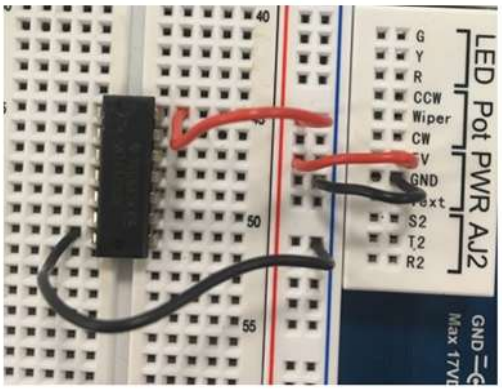


Figure 14: How to supply power to SN74LS32N

- Connect pin 1 and 2 to 0 V.
- Make sure that the NI myDAQ is connected correctly to your computer. This is explained in more detail in the previous section.
- Ensure that the 5V switch on the breadboard is at the 'ON' position.
- Open the NI ELVISmx Instrument Launcher and double click on the Digital Multimeter.
- Choose DC Voltage under Measurement Settings on the DMM.
- Run the DMM.
- Measure the voltage on the pins 7, 14, 1, 2, 3 to the nearest mV and record the measurement in the table given below.

Pin 7	Pin 14	Pin 1	Pin 2	Pin 3
0.100	4800	0	0	0

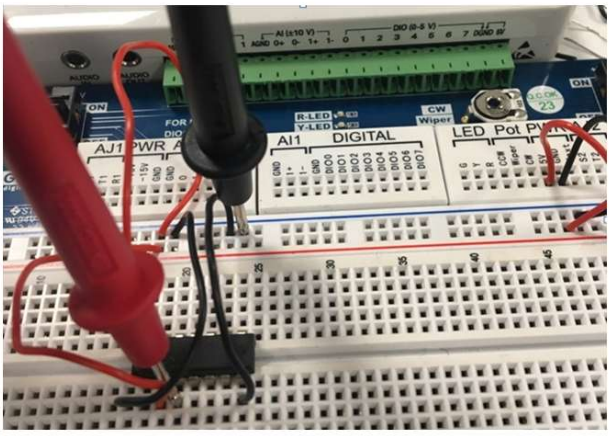


Figure 15: Measuring the voltage on pin 3 of SN74LS32N using the DMM probes. Note that the red probe is touching the output pin (pin 3) and the black probe (the reference probe) is touching the ground rail as the voltage is measured with respect to Ground.

Task 1

(8 marks)

1. Now change the voltages on the input pins of OR gate 1 of the SN74LS32N according to table given below and record the voltages on pin 1, 2 and 3 to the nearest mV in the same table. You could get negative voltages. If this is the case, write down the magnitude of the voltage.

Voltage Inputs		Recorded voltage (to nearest mV)		
Pin 1	Pin 2	Pin 1	Pin 2	Pin 3
0 V	0 V	0	0	0
0 V	5 V	0	4770	4230
5 V	0 V	4780	0	4220
5 V	5 V	4780	4780	4230

2. By considering the voltages that you have recorded, **briefly explain** how your results suggest the behaviour of an OR gate.

Represents truth table of an OR gate where there is an output voltage in pin 3 when there is voltage present in either one of the inputs (pin 1 or pin 2) or both.

Part B: Exploring Valid Inputs: How low is high and how high is low?

In the previous section, you used inputs that are directly taken from the 5V power supply of the myDAQ. However, in reality the voltage output from the power supply is not exactly 5 V; often it is below 5 V and the voltage output fluctuates. In other words, actual voltage signals are 'noisy' and fluctuate around an average value.

This section explores which input voltage range can be considered as logic LOW and which input voltage range can be considered as logic HIGH. The voltage input to one of the input pins of the OR gate 1 of the SN74LS32N chip is varied using equal steps from 0 V to 5 V and the resulting voltage outputs should be recorded.

For this purpose, a Variable Power Supply (VPS) or a DC Sweep function would be ideal. However, the NI ELVISmx Instrument Launcher does not support such a feature for the NI myDAQ. Therefore, you will be using an external LabVIEW VI (Virtual Instrument - piece of software) in this part of the exercise.

Important: You should double check that you have connected the correct pins to the power supply before powering on the myDAQ. Otherwise, you could potentially damage the chips.

- Connect pin 7 of SN74LS32N to Ground, pin 14 to 5 V (provides power to chip). Then connect pin 1 to 0 V (Ground).
- Connect pin 2 of SN74LS32N to Analog Output channel 0 (AO0). The voltage input to pin 2 is varied by the LabVIEW program.
- Connect pin 3 of SN74LS32N to Analog input channel 0 (AI0+). Connect AI0- line to Ground. The output voltage on pin 3 is recorded by the LabVIEW program. The recorded data is exported to Excel.

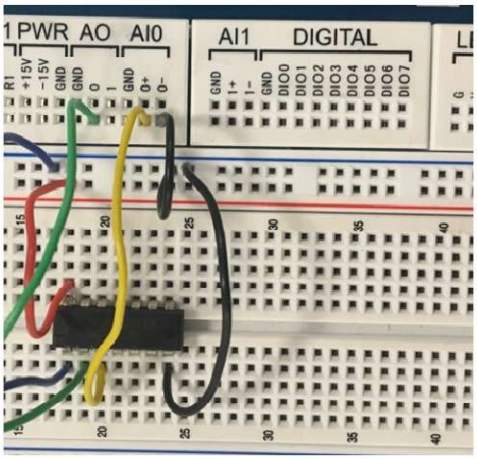


Figure 16: Connections to NI myDAQ for the DC Sweep

- Make sure your myDAQ is connected correctly to your computer.
- Make sure the name of the device is set as ‘myDAQ1’. If the name of the device is anything other than this, the LabVIEW program will return an error. The procedure for this is mentioned in the Getting Started section of this lab.
- Open the LabVIEW VI called “DC Sweep”.
- Press the ‘run’ button in the toolbar to start the VI.

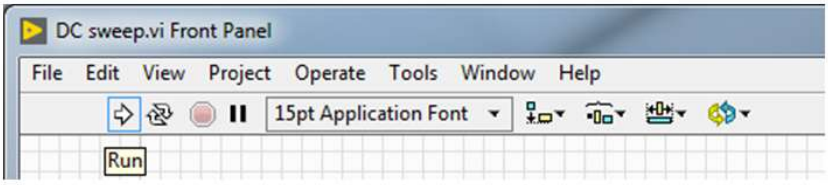


Figure 17: Run button on the toolbar

(Note: the LabVIEW VI called “DC Sweep” has been provided for you in Blackboard under EEEN10131 Digital Systems Design. You will need to download the VI, “DC Sweep”, from Blackboard to your personal computer and the university’s computers you are using).

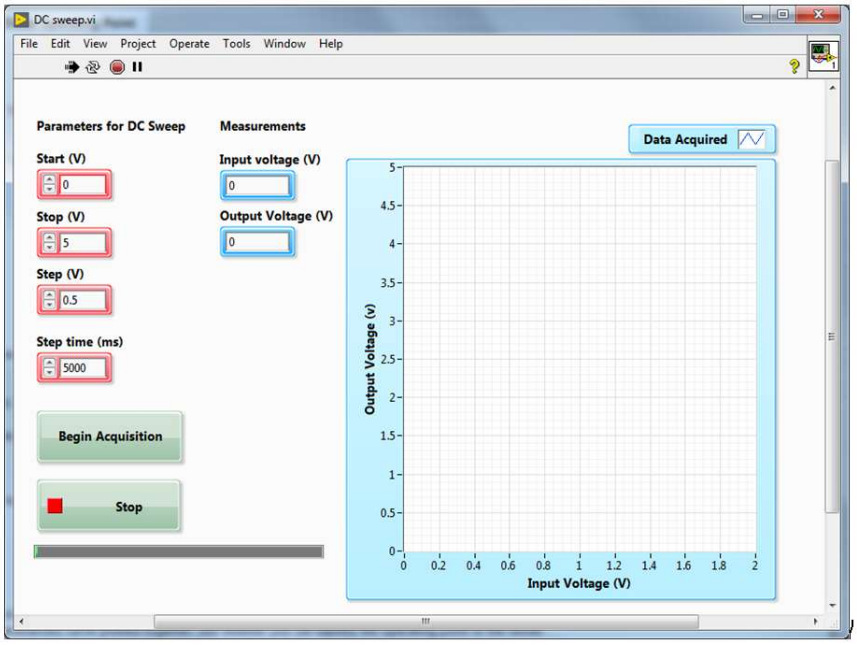


Figure 18: What you should see after you press Run button

- Enter the following parameters to the controls on the LabVIEW VI.
    - Start Voltage (V) : 0
    - Stop Voltage (V) : 5
    - Step (V) : 0.5
    - Step Interval (ms): 5000

Important: Do not experiment with Stop Voltages higher than 5 V. This could lead to high current draw from the Analog outputs of the myDAQ which could be potentially harmful to the myDAQ.

- Press ‘Begin Acquisition’ button.
- The Output Voltage and Input Voltage indicators will update every 5 seconds (see Figure 19 on the next page). Record these readings in the table given on the next page to the nearest mV. Don’t panic if you miss one point. Wait till the DC sweep reaches the stop voltage. Then re-run the program by entering all the required parameters and pressing ‘Begin Acquisition’ button again. The graph of the Output voltage against input voltage will be plotted automatically.

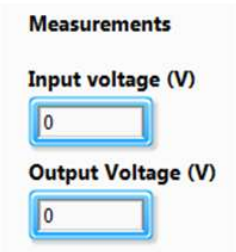


Figure 19: Input Voltage and Output Voltage Indicators. These will update every 5 seconds.

- Record the readings in the table given below.

Input (V)	Output(V)		Input (V)	Output(V)
			2.5	4.328
0	0.105		3	4.326
0.5	0.105		3.5	4.327
1	0.117		4	4.324
1.5	4.323		4.5	4.326
2	4.323		5	4.324

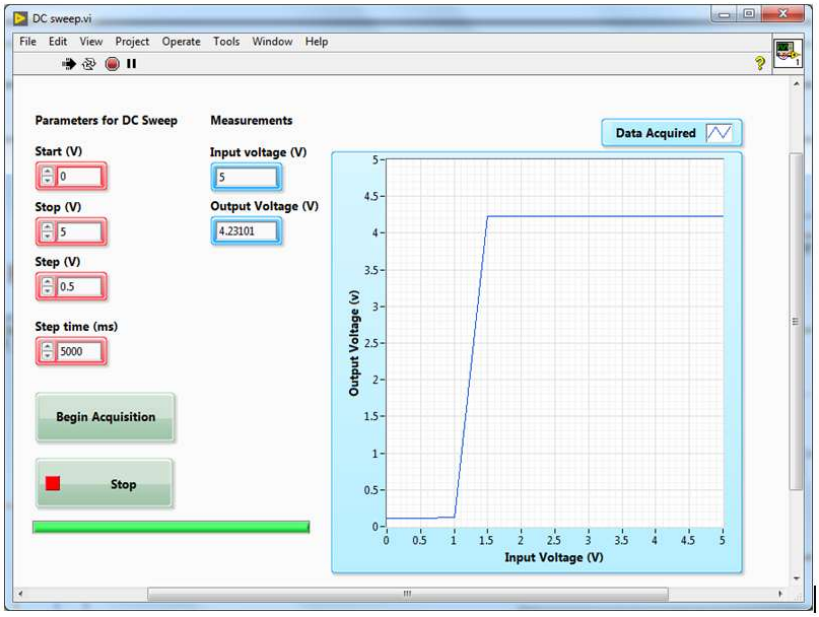
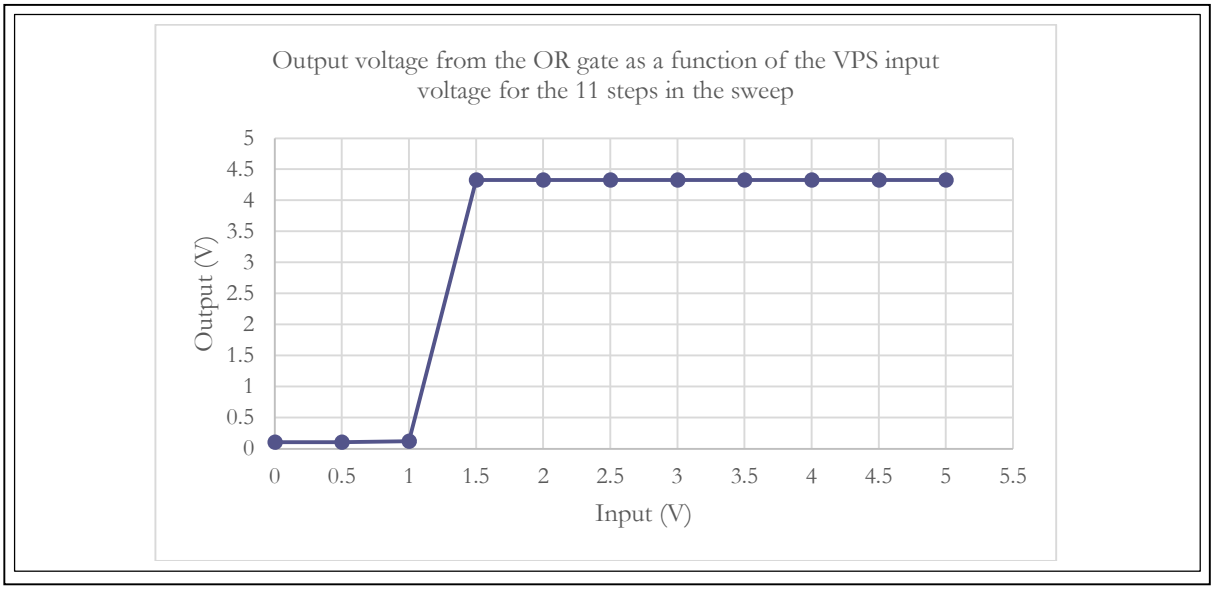


Figure 20: Results of the DC Sweep experiment

Task 2

(6 marks)

Plot the output voltage from the OR gate as a function of the VPS input voltage for the 11 steps in the sweep. Clearly show each data point. As for all graphs, you must label the axes. You can use excel to plot the graph.



Task 3

(4 marks)

The aim of this task is to gain improved understanding of the transition region where the output changes from low to high. You should now improve your results in and around this transition region where the output is very sensitive to the input. Change the following parameters in the LabVIEW DC Sweep program.

- Start Voltage (V) : 0.9*
- Stop Voltage (V) : 1.2*
- Step (V) : 0.03
- Step Interval (ms): 5000

*Note: you may have to widen the Start/Stop range to ensure that the output voltage starts at 0V and rises to 5V.

Record the output voltages to the nearest mV in the table given below.

Input (V)	Output(V)		Input (V)	Output(V)		Input (V)	Output(V)		Input (V)	Output(V)
0.90	0.106		1.02	1.071		1.14	1.248			
0.93	0.108		1.05	1.111		1.17	1.293			
0.96	0.112		1.08	1.155		1.20	1.340			
0.99	0.116		1.11	1.200						

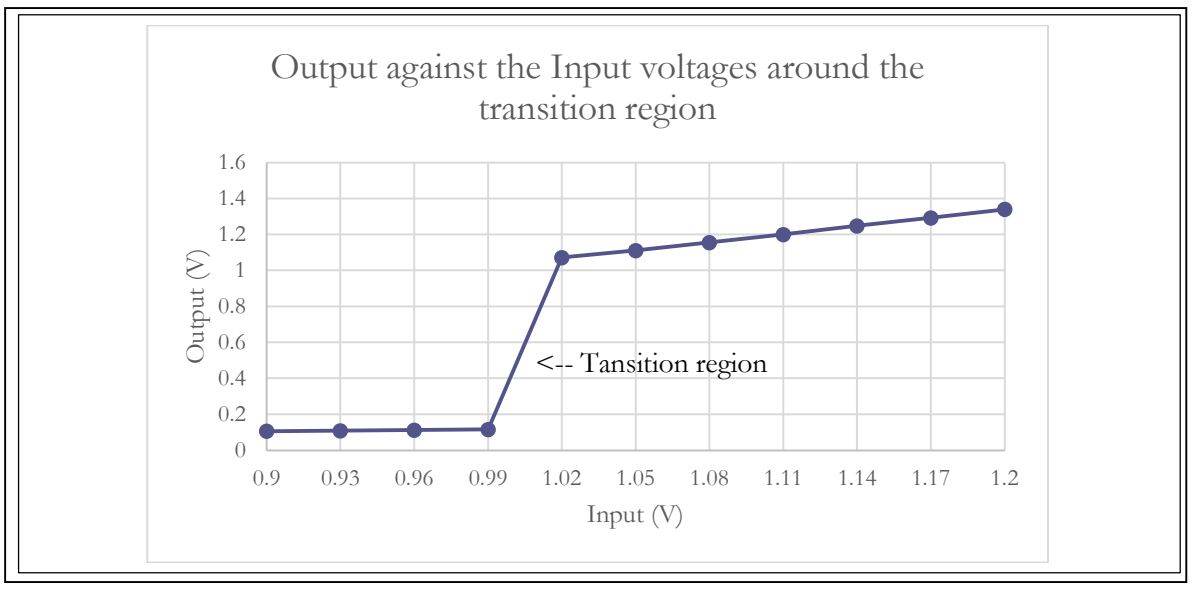
After you have recorded all the values correctly, press the 'Stop' button and then exit the LabVIEW program.

Task 4

(16 marks)

From the recorded values in Task 3 plot the measured output against the input voltages around the transition region. The horizontal axis should cover the range 0.9 V to 1.2 V (or the extended range, if required). You should show all the data points accompanied by the corresponding output voltage to the nearest mV. You might notice that the graph plotted in the program has sharp kinks on it. These happen due to outliers in the data acquired. When you plot the graph you should make the curve smooth so that you can clearly show the transition region. You may use excel to plot the graph.

You should indicate where the transition region occurs.



Task 5

(4 marks)

From your results in Task 4, answer the following questions. (2 marks each)

1) Suggest what range of input voltages would produce an output logic LOW?

Vin ≤ 0.99V

2) Suggest what range of input voltages would produce an output logic HIGH?

Vout ≥ 1.02V

Part C: Behaviour of 2-input NAND gate using SN74AC00N

In this part of the lab we will be using SN74AC00N quad (4 gates) 2-input NAND gate chip by Texas Instruments. This uses CMOS technology as suggested by “AC” in the name. CMOS is now the most common digital technology.

The datasheet of this integrated circuit (chip) is given in the folder of this laboratory on Blackboard. Have a look at the datasheet to learn more about the recommended operating conditions and the schematic diagram of the integrated circuit.

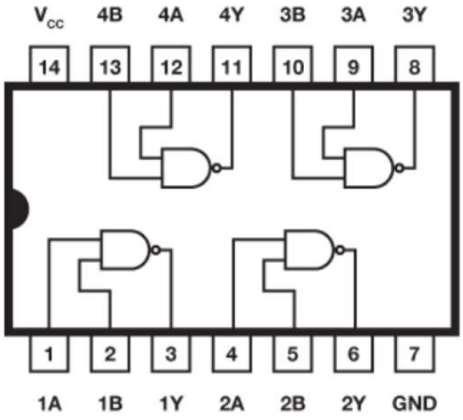


Figure 21: Pinout of SN74AC00N (image source: datasheetdir.com)

As you will have probably realised from Part A, it is inconvenient to have to change the wiring and probe pins with the DMM each time you have to change an input. The NI myDAQ provides a number of digital I/O channels to make this process easier.

In this exercise we will use the NAND gate number 4 on the chip.

- Place the chip on the breadboard. Connect pin 14 (Vcc) to 5 V and pin 7 to Ground to supply power to the chip.
- Connect pin 13 to DIO0, pin 12 to DIO1 and pin 11 to DIO 7.
- Make sure that your myDAQ is still connected to your computer.
- Make sure 5 V switch on the breadboard is at the 'ON' position.
- Open Digital Reader and Digital Writer from the NI ELVISmx Instrument Launcher.
- Configure digital lines as inputs to the myDAQ (outputs from the circuit you have built) by choosing the required group of lines in the Digital Reader. Output from the NAND gate is connected to DIO7.Hence, choose 'Lines to Read' as 4-7 in the Digital Reader.
- Configure digital lines outputs from the myDAQ (inputs to the circuit you have built) by choosing the required group of lines in the Digital Writer. Inputs to the NAND gate are connected to DIO0 and DIO1. Hence, choose 'Lines to Write' as 0-3 in the Digital Writer.

  Important: You cannot configure one line as both an input and output simultaneously as this will cause a conflict between the Digital Reader and Digital Writer.
- Run both the Digital Reader and the Digital Writer.
- Set all lines on the Digital Writer to logic low ('LO').
- Exercise all the possible combinations on the Digital Writer for lines 0 and 1 and record the output from the Digital Reader on line 7. When a line lights up on the Digital Reader that means its value is logic high.

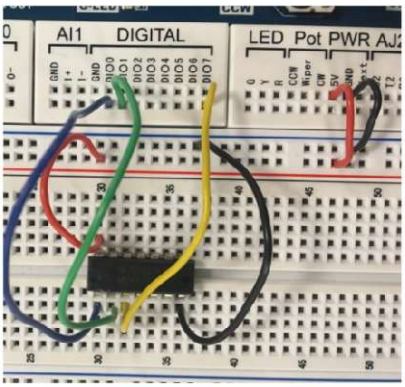


Figure 22: Connections to NI myDAQ

Task 6

(6 marks)

From your results, produce a truth table for this gate.

Inputs to NAND gate		Output from NAND gate
Pin 13	Pin 12	Pin 11
0	0	1
1	0	1
0	1	1
1	1	0

Briefly explain how the truth table you've produced confirms that the SN74AC00N provides the NAND function.

The output from a NAND gate is the opposite of the output of an AND gate. Where the inputs to an AND gate are (0,0), (1,0), (0,1),(1,1) the outputs are 0,0,0,1 respectfully, therefore the output for an NAND gate where the inputs are (0,0), (1,0), (0,1),(1,1) would be the opposite i.e. 1,1,1,0 respectfully therefore showing SN74AC00N provides the NAND function

Task 7

This task is based on designing a Circuit to implement a sum of products function on the NI myDAQ. Refer back to the example provided at the Essential Lab Preparation section if you need more help with this task.

Below is a truth table for a 3-input combinatorial circuit that you are required to build. (Use the correct truth table for you, based on your surname):

Surname from A to G

A	B	C	Z
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

Surname from H to Q

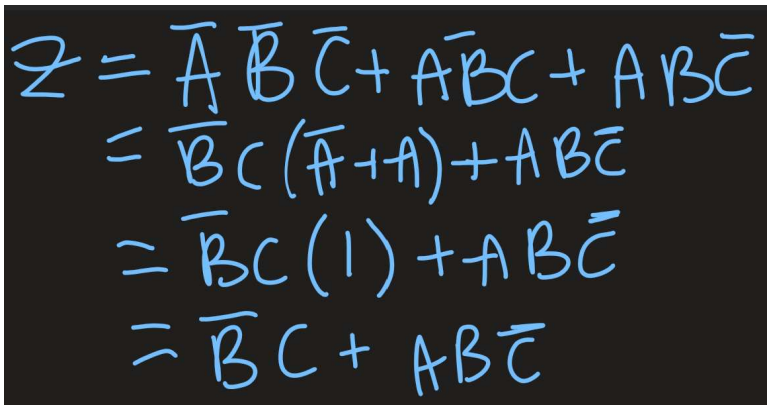
A	B	C	Z
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Surname from R to Z

A	B	C	Z
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

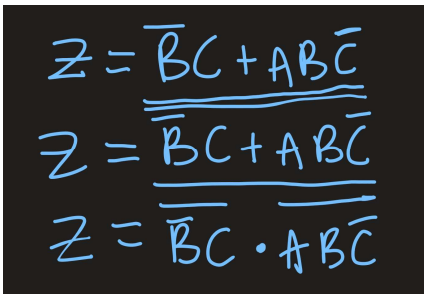
Task 7.1
(6 marks)

Deduce the canonical sum of products expression and then find the minimised sum of products expression.



Task 7.2
(6 marks)

Convert the minimised expression for Z from Task 7.1 to NAND-NAND format.



Task 7.3
(16 marks)

Draw the logic circuit corresponding to the minimised NAND-NAND function you obtained in Task 7.2 using SN74AC00N (which has 4 x 2 input NAND gates) and 74HC20N (which has 2 x 4 input NAND gates). As mentioned in the example at the start of the lab, this diagram should give all the information that is necessary should another person wish to build it on the NI myDAQ without reference to any other documents. You will lose marks if any information is not supplied.

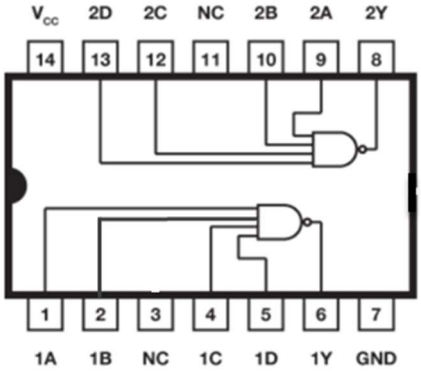


Figure 23: pinout details of 74HC20N by NXP Semiconductors (image source: datasheetdir.com)

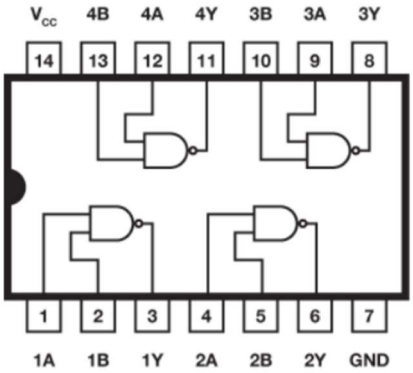
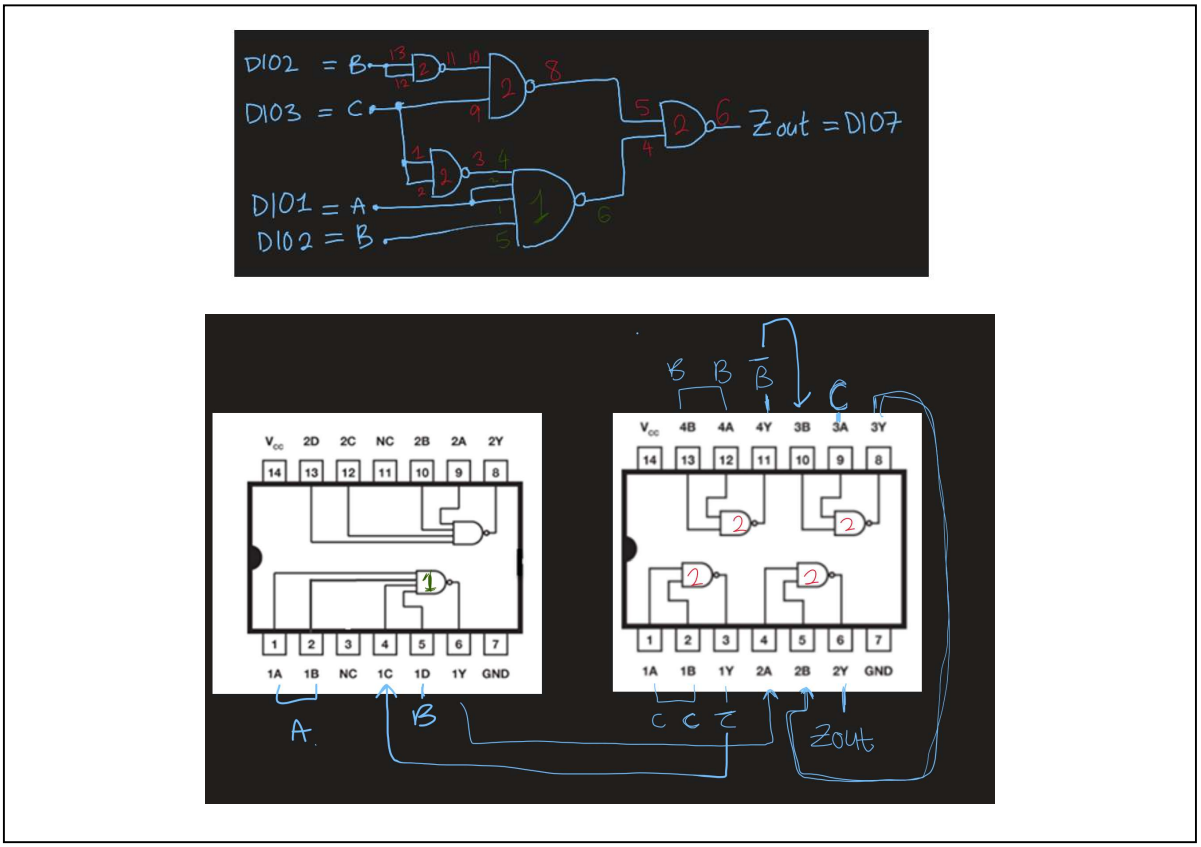


Figure 24: pinout details of SN74AC00N (image source: datasheetdir.com)



Task 7.4
(16 marks)

Build the logic circuit from the diagram you drew in Task 7.3 on your breadboard and confirm that it is behaving correctly. To do this, Write all the input combinations for A, B and C using the Digital Writer and Read the output Z from the Digital Reader and confirm that the output is correct for all the input combinations.

Attend the scheduled on-campus lab at allocated one-hour time slot, and demonstrate the circuit built with the connection on MyProto board and MyDaq (you may assemble the circuit before attending or build the circuit on-site within the time slot). The mark is given at the time of demonstration, which is:_____/16 .

You are only required to attend at your allocated one-hour time slot within your timetable's period.

Surname from A to G: 1st hour of the 3-hour period

Surname from H to Q: 2nd hour of the 3-hour period

Surname from R to Z: 3rd hour of the 3-hour period

Task 8
(12 marks)

Using the minimised sum of products expression for Z that you derived in Task 7.1

i) Find a logic expression for Z using only NOR gates.
ii) Draw the circuit diagram for this expression using only 2-input NOR gates and 4-input NOR gates. You do not need to show any pin numbers or chip identification.

